

EDEXCEL INTERNATIONAL A LEVEL

WME01/01 January 2026 Worked Solutions

January 2026 paper rebuilt with the worked solution after each question.

7

paper questions

WME01

Mechanics 1

**Single-
paper
solutions**standalone IAL
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PAST PAPER

WME01/01 January 2026

January 2026 | 7 questions | 75 marks

7

questions

75

marks

Answers

worked solution
after each
question

Question 1

Momentum, Impulse & Collisions

1. A particle P of mass $3m$ and a particle Q of mass $5m$ are on a smooth horizontal surface. The particles move towards each other in opposite directions along the same straight line and collide.

Immediately before the collision, the speed of P is $3u$ and the speed of Q is $4u$

The magnitude of the impulse exerted on P by Q in the collision is $\frac{33}{2}mu$

Find

- (i) the speed of P immediately after the collision,
- (ii) the speed of Q immediately after the collision.

(6)

Worked Solution - Question 1

Topic group

1. Choose a sign convention

Take the initial direction of P as positive. Then before impact, $v_P = 3u$ and $v_Q = -4u$. The impulse on P acts opposite to P's initial motion, so its signed impulse is $-\frac{33}{2}mu$.

2. Use impulse on P

For P, impulse equals change in momentum: $-\frac{33}{2}mu = 3m(v_P - 3u)$. Dividing by $3m$ gives $v_P - 3u = -\frac{11u}{2}$, so $v_P = -\frac{5u}{2}$.

3. Interpret the speed of P

The negative sign means P has reversed direction. Its speed is therefore $\left|-\frac{5u}{2}\right| = \frac{5u}{2}$.

4. Use the equal and opposite impulse on Q

Q receives impulse $+\frac{33}{2}mu$ in the positive direction. Hence $\frac{33}{2}mu = 5m(v_Q - (-4u)) = 5m(v_Q + 4u)$.

5. Interpret the speed of Q

Dividing by $5m$ gives $v_Q + 4u = \frac{33u}{10}$, so $v_Q = -\frac{7u}{10}$. Q is still moving in its original direction, and its speed is $\frac{7u}{10}$.

Final answer

(i) speed of P = $\frac{5u}{2}$. (ii) speed of Q = $\frac{7u}{10}$.

Question 2

Kinematics Graphs

2.

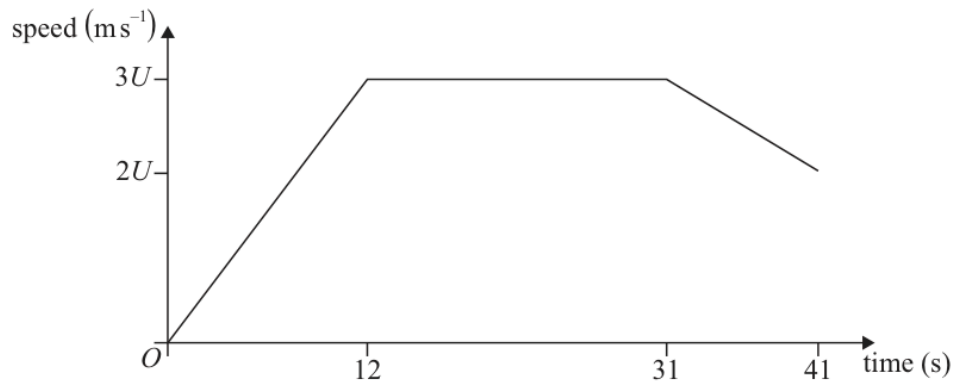


Figure 1

A particle P starts from rest at time $t = 0$ and moves in a straight line

- from $t = 0$ to $t = 12$ s, P accelerates uniformly until it reaches a speed of $3U \text{ m s}^{-1}$
- from $t = 12$ s to $t = 31$ s, P moves with constant speed $3U \text{ m s}^{-1}$
- from $t = 31$ s to $t = 41$ s, P decelerates uniformly until it has speed $2U \text{ m s}^{-1}$

as shown on the speed-time graph in Figure 1.

The distance travelled by P between $t = 0$ and $t = 41$ s is 600 m.

- Show that $U = 6$ (3)
- Find the distance travelled by P whilst moving from rest to a speed of 6 m s^{-1} (2)
- Find the acceleration of P between $t = 0$ and $t = 12$ s. (1)
- Find the deceleration of P between $t = 31$ s and $t = 41$ s. (2)
- Sketch an acceleration-time graph to represent the motion of P from $t = 0$ to $t = 41$ s. (2)

Worked Solution - Question 2

Topic group

1. Use area under the speed-time graph

Distance is the area under a speed-time graph. The three areas are a triangle from 0 to 12, a rectangle from 12 to 31, and a trapezium from 31 to 41.

2. Show that U is 6

$\frac{1}{2}(12)(3U) + 19(3U) + \frac{1}{2}(3U + 2U)(10) = 600$. This gives
 $18U + 57U + 25U = 600$, so $100U = 600$ and $U = 6$.

3. Find the distance to speed 6

Since $U = 6$, the final speed 6 m s^{-1} occurs during the first section. The acceleration there is $\frac{18 - 0}{12} = 1.5 \text{ m s}^{-2}$, so the time to reach 6 is 4 s. The distance is $\frac{1}{2}(4)(6) = 12 \text{ m}$.

4. Find the acceleration and deceleration

From 0 to 12, $a = \frac{18}{12} = 1.5 \text{ m s}^{-2}$. From 31 to 41, the speed falls from 18 to 12, so the deceleration is $\frac{18 - 12}{10} = 0.6 \text{ m s}^{-2}$.

5. Describe the acceleration-time graph

The acceleration-time graph has three horizontal sections: 1.5 from $0 \leq t \leq 12$, 0 from $12 < t \leq 31$, and -0.6 from $31 < t \leq 41$.

Final answer

(a) $U = 6$. (b) 12 m . (c) 1.5 m s^{-2} . (d) 0.6 m s^{-2} . (e) $a(t) = 1.5, 0, -0.6$ on the three intervals.

Question 3

Forces

3. A particle moves on a smooth horizontal plane under the action of three horizontal forces, $\mathbf{F}_1$, $\mathbf{F}_2$ and $\mathbf{F}_3$, where

$$\mathbf{F}_1 = (2p\mathbf{i} - 3p\mathbf{j})\text{N}$$

$$\mathbf{F}_2 = (q\mathbf{i} + \mathbf{j})\text{N}$$

$$\mathbf{F}_3 = (3\mathbf{i} - q\mathbf{j})\text{N}$$

where p and q are constants.

The resultant force acts in the direction of the vector $(4\mathbf{i} - 5\mathbf{j})$

- (a) Show that

$$2p - q = 19 \quad (4)$$

The mass of the particle is 0.5 kg.

Given that $p = 7$

- (b) find the acceleration of the particle. (4)

Worked Solution - Question 3

1. Add the three forces

The resultant force is

$$(2p\mathbf{i} - 3p\mathbf{j}) + (q\mathbf{i} + \mathbf{j}) + (3\mathbf{i} - q\mathbf{j}) = (2p + q + 3)\mathbf{i} + (-3p + 1 - q)\mathbf{j}.$$

2. Use the direction vector

The resultant acts in the direction $4\mathbf{i} - 5\mathbf{j}$, so the ratio of the components is the

same: $\frac{-3p + 1 - q}{2p + q + 3} = \frac{-5}{4}.$

3. Show the required relation

Cross-multiplying gives $4(-3p + 1 - q) = -5(2p + q + 3)$. Therefore $-12p + 4 - 4q = -10p - 5q - 15$, so $2p - q = 19$.

4. Find q when p is 7

With $p = 7$, the relation gives $14 - q = 19$, hence $q = -5$.

5. Apply Newtons second law

Substitute $p = 7$ and $q = -5$ into the resultant: $\mathbf{F} = 12\mathbf{i} - 15\mathbf{j}$. Since the mass is 0.5 kg , $\mathbf{F} = m\mathbf{a}$ gives $12\mathbf{i} - 15\mathbf{j} = 0.5\mathbf{a}$.

6. Find the acceleration

Dividing by 0.5 gives $\mathbf{a} = 24\mathbf{i} - 30\mathbf{j} \text{ m s}^{-2}$.

Final answer

(a) $2p - q = 19$. (b) $\mathbf{a} = 24\mathbf{i} - 30\mathbf{j} \text{ m s}^{-2}$.

Question 4

Moments

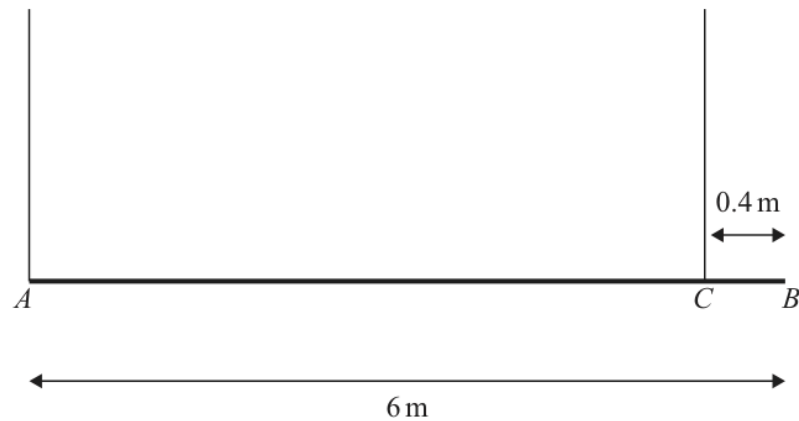


Figure 2

Figure 2 shows a uniform rod AB of length 6 m suspended by two light vertical ropes.

The first rope is attached to the rod at A .

The second rope is attached to the rod at the point C , where $CB = 0.4$ m.

The rod has mass 30 kg.

A particle of mass 20 kg is now attached to the rod at B .

The rod is in equilibrium in a horizontal position.

(a) Find

- (i) the tension in the rope attached to the rod at C ,
- (ii) the tension in the rope attached to the rod at A .

(6)

The particle of mass 20 kg at B is removed and replaced by a particle of mass M kg.

The rod remains in equilibrium in a horizontal position.

(b) Find the exact maximum value of M .

(3)

Worked Solution - Question 4

1. Set the distances

The rod is 6 m long and C is 0.4 m from B, so $AC = 5.6$ m. The rod is uniform, so its weight $30g$ acts at the midpoint, 3 m from A. The 20 kg particle at B has weight $20g$.

2. Take moments about A

Moments about A remove the tension at A. Clockwise moments from the rod and particle equal the anticlockwise moment from the rope at C:

$$T_C(5.6) = 30g(3) + 20g(6).$$

3. Find the tension at C

$T_C(5.6) = 210g$, so $T_C = 37.5g = 367.5$ N. To suitable accuracy, $T_C = 368$ N, or 370 N to 2 significant figures.

4. Use vertical equilibrium

Vertically, $T_A + T_C = 30g + 20g = 50g$. Therefore

$T_A = 50g - 37.5g = 12.5g = 122.5$ N, so $T_A = 123$ N, or 120 N to 2 significant figures.

5. Use the limiting case for maximum M

For the greatest possible mass at B, the rope at A is just slack, so $T_A = 0$. Take moments about C: the clockwise moment of Mg at B balances the anticlockwise moment of the rod weight.

6. Solve for M

The distance from C to B is 0.4 m and the rod weight acts $5.6 - 3 = 2.6$ m to the left of C. Hence $Mg(0.4) = 30g(2.6)$, so $M = \frac{78}{0.4} = 195$.

Final answer

$(a)(i) T_C = 368 \text{ N or } 370 \text{ N.}$ $(a)(ii) T_A = 123 \text{ N or } 120 \text{ N.}$ $(b) M = 195$

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Question 5

Resolving Forces, Inclined Planes

5. The points O , A , B and C are on a rough horizontal surface where $OABC$ is a straight line.

A particle is projected horizontally from O and slides across the surface, passing through A , B and C in that order.

The particle passes through

- A with speed $U \text{ ms}^{-1}$
- B with speed $\frac{U}{2} \text{ ms}^{-1}$

The coefficient of friction between the particle and the surface is $\frac{1}{7}$

It takes 0.75 s for the particle to move from A to B .

- (a) Show that $U = 2.1$ to 2 significant figures.

(6)

The particle passes through C with speed $\frac{U}{3} \text{ ms}^{-1}$

- (b) Find the time taken to move from B to C .

(3)

- (c) Find the distance AC .

(3)

Worked Solution - Question 5

Topic group

1. Find the acceleration from friction

On a horizontal surface, $R = mg$ and the friction is $F = \mu R = \frac{1}{7}mg$. The friction is the only horizontal force, so the acceleration is opposite the motion with magnitude $\frac{F}{m} = \frac{g}{7} = 1.4 \text{ m s}^{-2}$.

2. Use A to B to find U

From A to B, $u = U$, $v = \frac{U}{2}$, $a = -\frac{g}{7}$ and $t = 0.75$. Using $v = u + at$:

$$\frac{U}{2} = U - \frac{g}{7}(0.75).$$

3. Show the value of U

With $g = 9.8$, $\frac{g}{7}(0.75) = 1.05$, so $\frac{U}{2} = U - 1.05$. Hence $U = 2.10 \text{ m s}^{-1}$, which is 2.1 m s^{-1} to 2 significant figures.

4. Find the time from B to C

From B to C, the speed falls from $\frac{U}{2}$ to $\frac{U}{3}$ with the same acceleration -1.4 . Thus

$$\frac{U}{3} = \frac{U}{2} - 1.4t.$$
 With $U = 2.1$, this gives $0.7 = 1.05 - 1.4t$, so $t = 0.25 \text{ s}$.

5. Find the distance AC

Use average speed on each section. $AB = \frac{U + U/2}{2}(0.75) = 1.18125 \text{ m}$ and
 $BC = \frac{U/2 + U/3}{2}(0.25) = 0.21875 \text{ m}$. Therefore
 $AC = 1.18125 + 0.21875 = 1.4 \text{ m}$.

Final answer

(a) $U = 2.1 \text{ m s}^{-1}$ to 2 s.f.. (b) 0.25 s. (c) $AC = 1.4 \text{ m}$.

Question 6

Working with Vectors

6. [In this question $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors directed due east and due north respectively and position vectors are given relative to a fixed origin O .]

At 9 am, a ship S leaves a harbour and moves with constant velocity $(10\mathbf{i} + 15\mathbf{j})\text{km h}^{-1}$

- (a) Find the exact speed of S

(2)

The harbour is at the origin O .

At 9 am, a boat B leaves the point with position vector $(50\mathbf{i} + 10\mathbf{j})\text{km}$ and moves with constant velocity $(-8\mathbf{i} + 10\mathbf{j})\text{km h}^{-1}$

- (b) Find the direction in which B is moving, giving your answer as a bearing to the nearest degree.

(3)

- (c) Show that at time t hours after 9 am,

$$\overrightarrow{SB} = [(50 - 18t)\mathbf{i} + (10 - 5t)\mathbf{j}]\text{km}$$

(4)

- (d) Show that B and S do not collide.

(3)

When B is north-east of S , the distance between S and B is $d\text{km}$.

- (e) Find the value of d .

(4)

Worked Solution - Question 6

1. Find the speed of S

The velocity of S is $10\mathbf{i} + 15\mathbf{j}$. Its speed is the magnitude:

$$\sqrt{10^2 + 15^2} = \sqrt{325} = 5\sqrt{13} \text{ km h}^{-1}.$$

2. Find the bearing of B

B has velocity $-8\mathbf{i} + 10\mathbf{j}$, so it moves west and north. The angle west of north satisfies $\tan \theta = \frac{8}{10}$, giving $\theta = 38.66^\circ$. The bearing is $360^\circ - 38.66^\circ = 321^\circ$ to the nearest degree.

3. Write both position vectors

At time t hours after 9 am, $\mathbf{r}_S = (10t)\mathbf{i} + (15t)\mathbf{j}$. Also

$$\mathbf{r}_B = (50\mathbf{i} + 10\mathbf{j}) + t(-8\mathbf{i} + 10\mathbf{j}) = (50 - 8t)\mathbf{i} + (10 + 10t)\mathbf{j}.$$

4. Find the relative position vector

$$\overrightarrow{SB} = \mathbf{r}_B - \mathbf{r}_S = [(50 - 8t) - 10t]\mathbf{i} + [(10 + 10t) - 15t]\mathbf{j} = (50 - 18t)\mathbf{i} + (10 - 5t)\mathbf{j}.$$

5. Show they do not collide

For a collision, both components of $\overrightarrow{SB}$ must be zero at the same time. From $50 - 18t = 0$, $t = \frac{25}{9}$. From $10 - 5t = 0$, $t = 2$. These are different, so B and S do not collide.

6. Find the north-east or south-west instant

When the separation is along a 45° diagonal, the two components have equal magnitude. Using the given condition, set $50 - 18t = 10 - 5t$, so $40 = 13t$ and $t = \frac{40}{13}$.

7. Find the distance d

At $t = \frac{40}{13}$, both components are $-\frac{70}{13}$. Therefore

$$d = \sqrt{\left(\frac{70}{13}\right)^2 + \left(\frac{70}{13}\right)^2} = \frac{70\sqrt{2}}{13} = 7.6 \text{ km to 2 significant figures.}$$

Final answer

(a) $5\sqrt{13} \text{ km h}^{-1}$. (b) 321° . (c) $\vec{SB} = (50 - 18t)\mathbf{i} + (10 - 5t)\mathbf{j}$. (d) no collision. (e) $d = \frac{70\sqrt{2}}{13} \text{ km} \approx 7.6 \text{ km}$.

Question 7

Resolving Forces, Inclined Planes

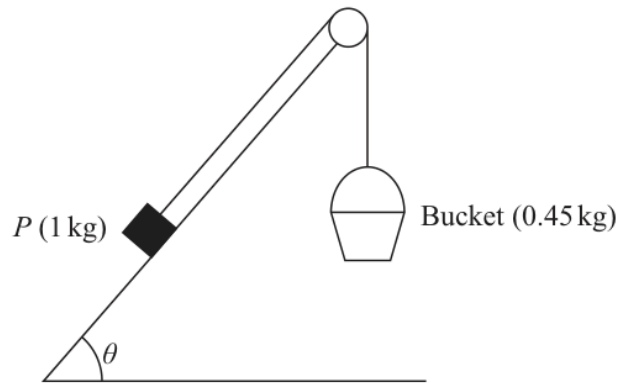


Figure 3

A rough plane is inclined at an angle θ to the horizontal, where $\tan \theta = \frac{4}{3}$

One end of a light inextensible string is attached to a package P .

The string passes over a smooth pulley that is fixed at the top of the plane.

The string from P to the pulley lies along a line of greatest slope of the plane.

The other end of the string is attached to a bucket.

The package P has mass 1 kg and is **held** at rest on the plane.

The bucket has mass 0.45 kg and hangs vertically below the pulley with the string taut, as shown in Figure 3.

The coefficient of friction between P and the plane is μ

When a small block of mass 0.2 kg is placed in the bucket, the system is released and remains at rest in **limiting equilibrium**, with P on the point of slipping down the plane.

- (a) Show that $\mu = \frac{1}{4}$ (7)

Additional identical blocks of mass 0.2 kg are added one at a time into the bucket until P starts to move **up** the plane.

At the instant when P starts to move up the plane, there are a **total** of n blocks in the bucket.

- (b) Find
- (i) the value of n
 - (ii) the magnitude of the initial acceleration of P
- (7)

Worked Solution - Question 7

Topic group

1. Find the trigonometric ratios

Since $\tan \theta = \frac{4}{3}$, use a 3-4-5 triangle: $\sin \theta = \frac{4}{5}$ and $\cos \theta = \frac{3}{5}$.

2. Use limiting equilibrium in part (a)

With one 0.2 kg block in the bucket, the bucket side has mass $0.45 + 0.2 = 0.65$ kg. Since the package is on the point of slipping down the plane, friction on P acts up the plane.

3. Find the friction force

The tension equals the bucket weight, so $T = 0.65g$. Along the plane for P in equilibrium, $T + F = g \sin \theta$. Hence $0.65g + F = \frac{4}{5}g$, so $F = \frac{3g}{20}$.

4. Find the normal reaction and μ

Perpendicular to the plane, $R = g \cos \theta = \frac{3}{5}g$. Since $F = \mu R$,

$$\mu = \frac{F}{R} = \frac{3g/20}{3g/5} = \frac{1}{4}.$$

5. Set the condition for upward motion

When P is about to move up the plane, friction acts down the plane. With n blocks in the bucket, the bucket-side mass is $0.45 + 0.2n$. The bucket weight must exceed the downslope resistance: $(0.45 + 0.2n)g > g \sin \theta + \mu R$.

6. Find n

Using $g \sin \theta = \frac{4}{5}g$ and $\mu R = \frac{1}{4} \cdot \frac{3}{5}g = \frac{3g}{20}$, the condition is
 $0.45 + 0.2n > \frac{4}{5} + \frac{3}{20} = 0.95$. Hence $0.2n > 0.50$, so $n > 2.5$ and the least integer is $n = 3$.

7. Apply Newtons second law after motion begins

For $n = 3$, the bucket-side mass is $0.45 + 0.6 = 1.05$ kg and the total moving mass is $1 + 1.05 = 2.05$ kg. The driving force is $1.05g - \frac{4}{5}g - \frac{3g}{20} = 0.10g$.

8. Find the acceleration

Using $F = ma$, $0.10g = 2.05a$, so $a = \frac{0.10g}{2.05} = \frac{2g}{41} = 0.478 \text{ m s}^{-2}$ to 3 significant figures.

Final answer

(a) $\mu = \frac{1}{4}$. (b)(i) $n = 3$. (b)(ii) $a = \frac{2g}{41} = 0.478 \text{ m s}^{-2}$ approximately